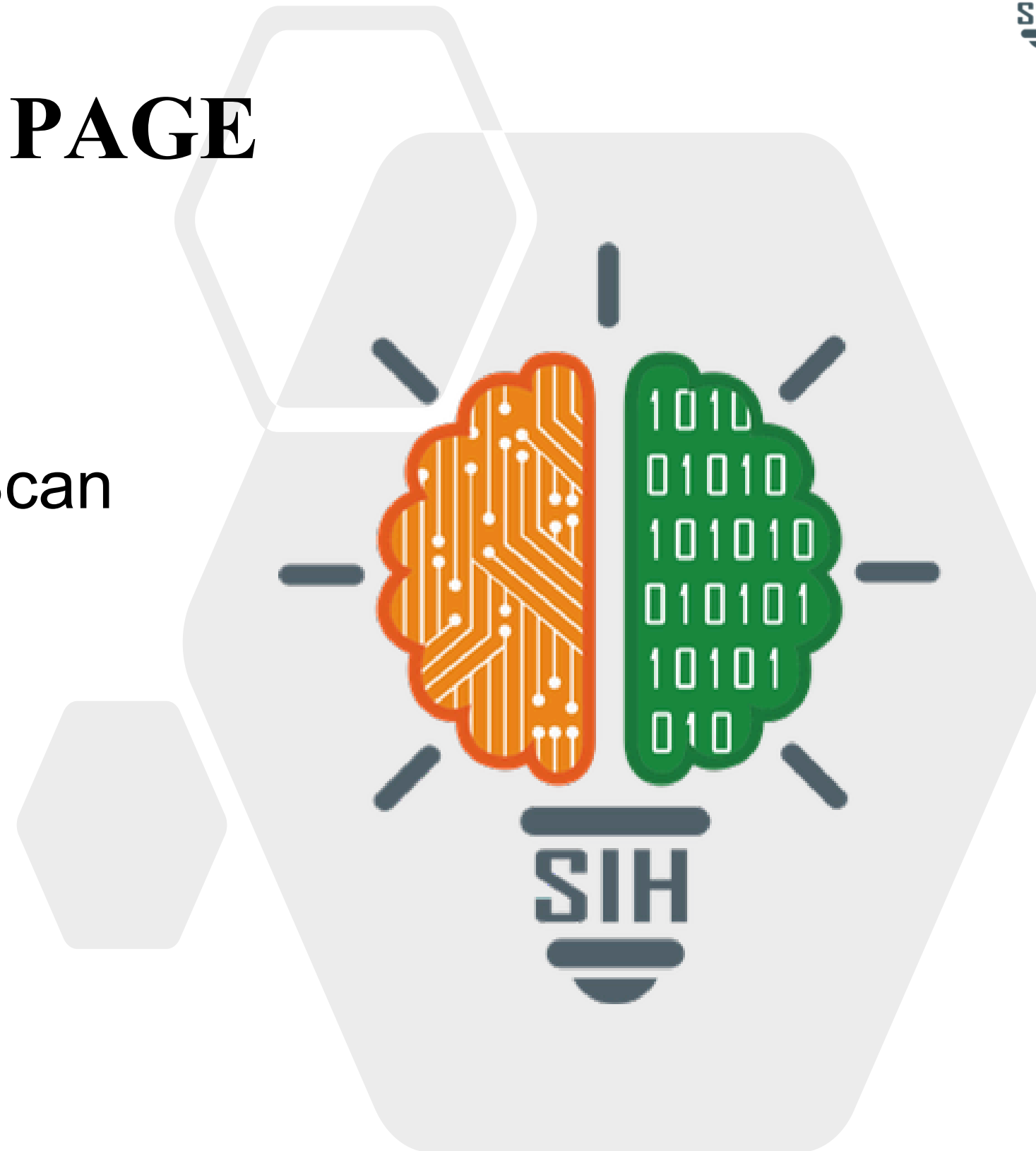


TITLE PAGE

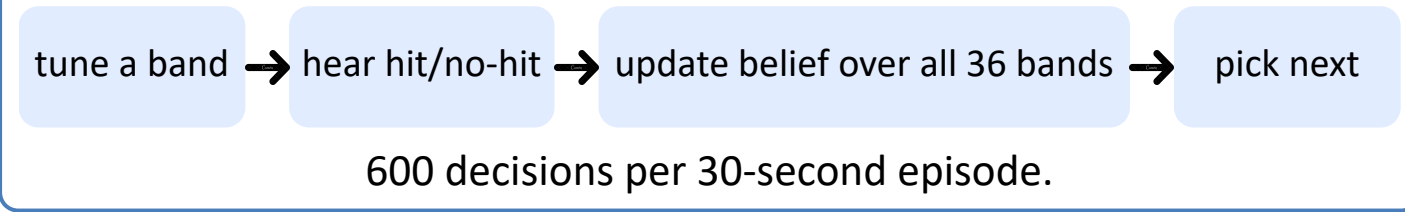
- **Problem Statement ID-** 26055
- **Problem Statement Title-** Smart Scan strategy for Electronic Warfare
- **Theme-** Robotics and Drones
- **PS Category-** Software
- **Team ID-** 124396
- **Team Name-** NIRVANA



A closed loop scheduler that beats every baseline including published state of the art literature

Today's ES receivers know how to sweep a fixed pre-mission schedule and cannot iterate. **NARADA closes this loop**

Closed loop mechanism



Reinforcement learning

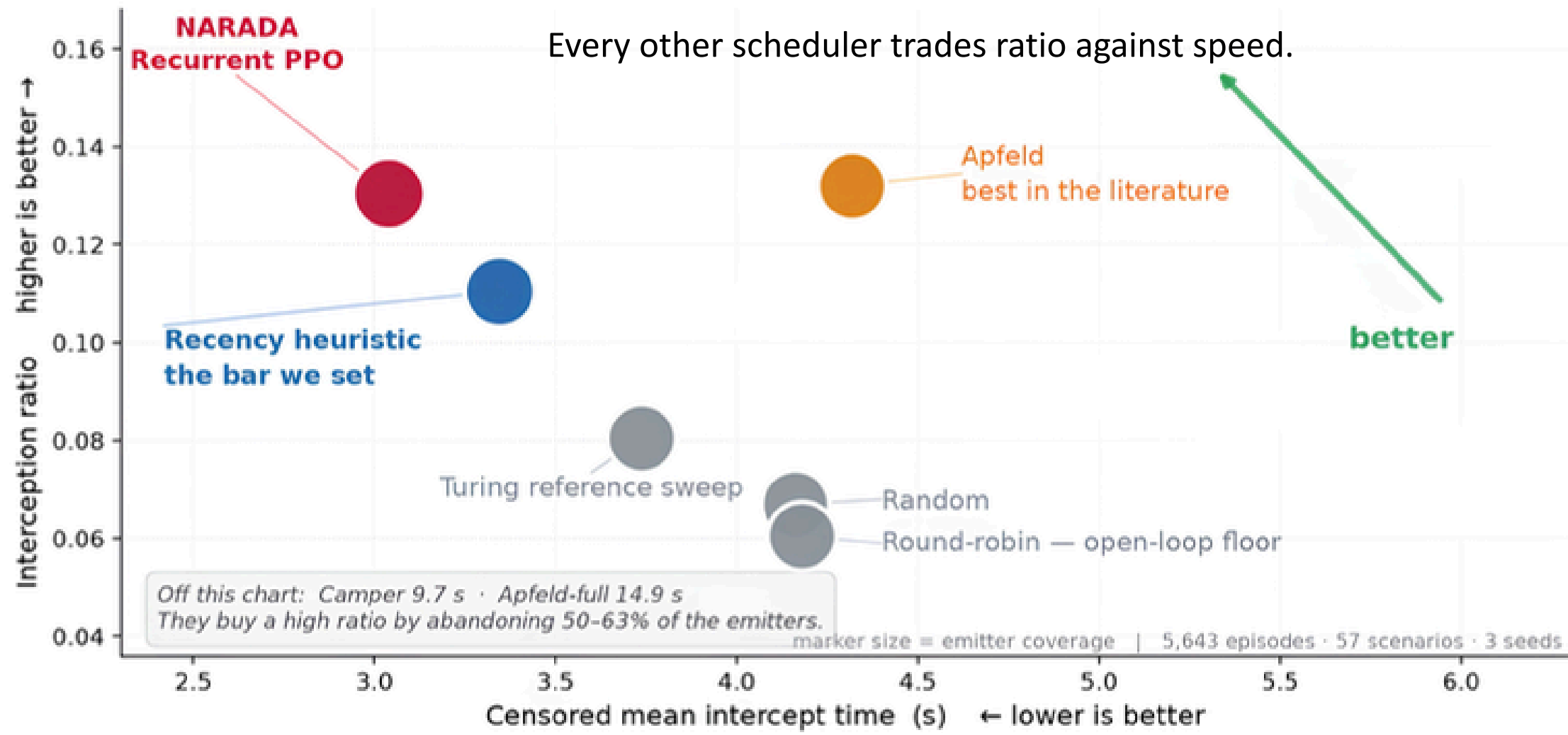
Recurrent PPO trained on a simulated RF environment built from 2.2 GB of real recorded radar data.

Adaptive Threat Library

Every ES system already carries a threat library. NARADA's scheduler can take a priority input from the environment .

METRIC	NARADA	OPEN LOOP SYSTEM	How much better?
Interception ratio	0.130	0.061	213%
Interception Time	3.04 s	4.18 s	27%
Coverage	0.902	0.865	104%

NARADA'S USP
ADAPTIVE THREAT LIBRARY
&
HUMAN ON THE LOOP
(not IN the loop)



Four layers of linear screening so NARADA'S scheduler's optimizes the 3 most important metrics

1. Scenario

2.2 GB of real recorded radar data that gives 3,443 reusable emitter contributions.
Every episode is fresh and new.

2. Truth

36 bands × 600 slots of 50 ms each. Ground truth of who's transmitting, independent of whether we look.

3. Receiver

Declares a hit if $\text{signal} + \text{noise} \geq \text{threshold}$.
Pd 0.84, Pfa 0.0013 are frozen.
The scheduler can aim the receiver, not improve it.

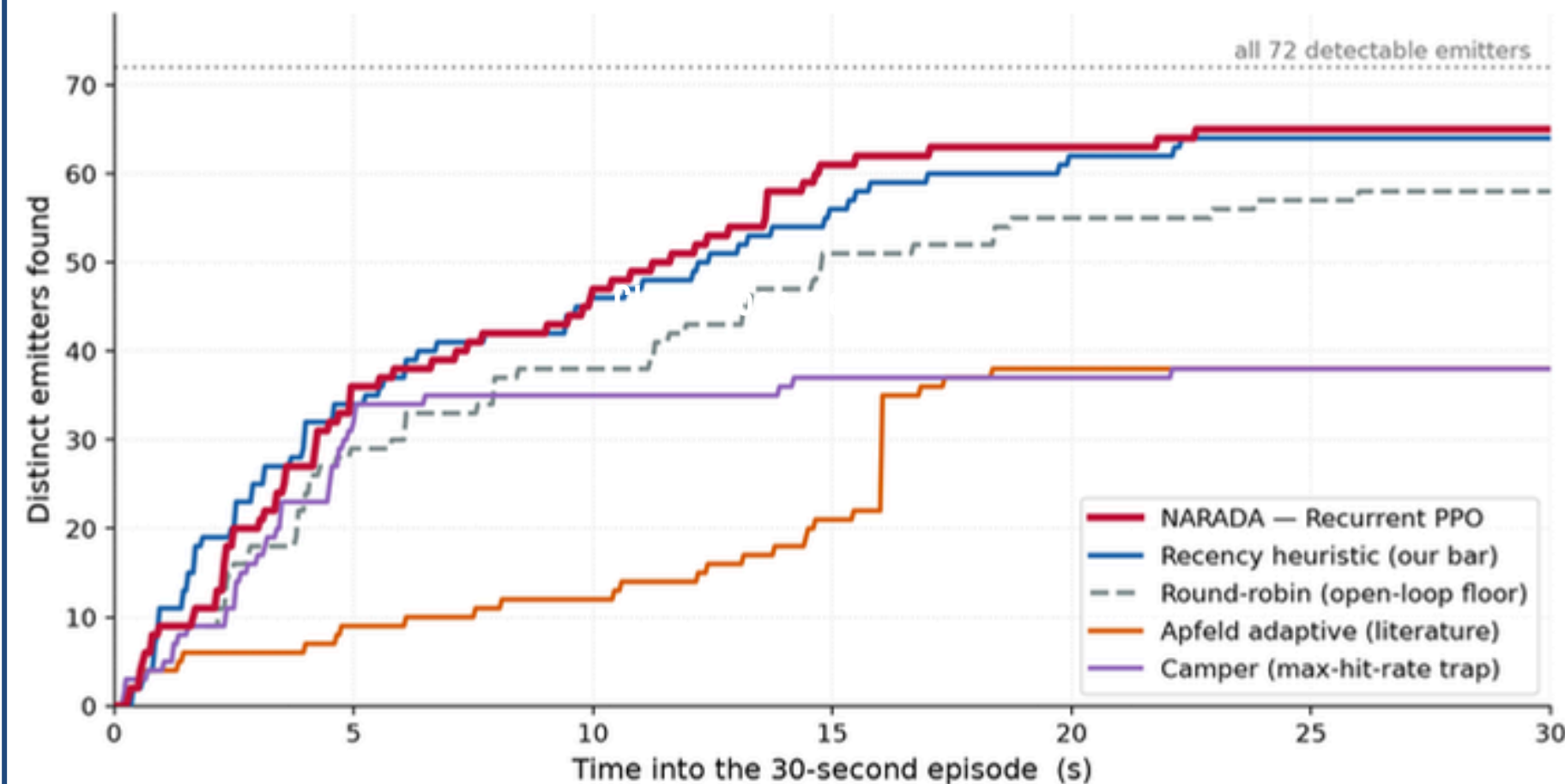
4. Scheduler

A Recurrent PPO that takes 36 actions, 183-value observation built only from its own scan history.
Ground Truth is invisible to it.

Python · Stable-Baselines3 · PyTorch · Gymnasium

Chosen because they are easily reproducible

In a 30 sec timeframe NARADA'S recurrent PPO was able to beat the Recency heuristic by at the 9 sec mark while staying almost parallel to it through out



Interception ratio
Intercept time
Coverage

We have to put
how to improve
the periodic
tracking here

Reward = Exploit what pays + Explore what's overdue – Cost of hogging airtime.

NARADA screens it against known good & bad policies before training, at 3.0σ separation.

A closed loop scheduler that beats every baseline including published state of the art literature

Today's ES receivers know how to sweep a fixed pre-mission schedule and cannot iterate. **NARADA** closes this loop

Closed loop mechanism

Tune a band → hear hit/no-hit → update belief over all 36 bands → pick next.
600 decisions per 30-second episode.

MAKE 4 BLOCKS FLOWCHART EXPRESSING THE FLOW OF THIS CLOSED LOOP

Reinforcement learning

Put the whole thing into a box
Recurrent PPO trained on a simulated RF environment built from .
2.2 GB of real recorded radar data

Adaptive Threat Library

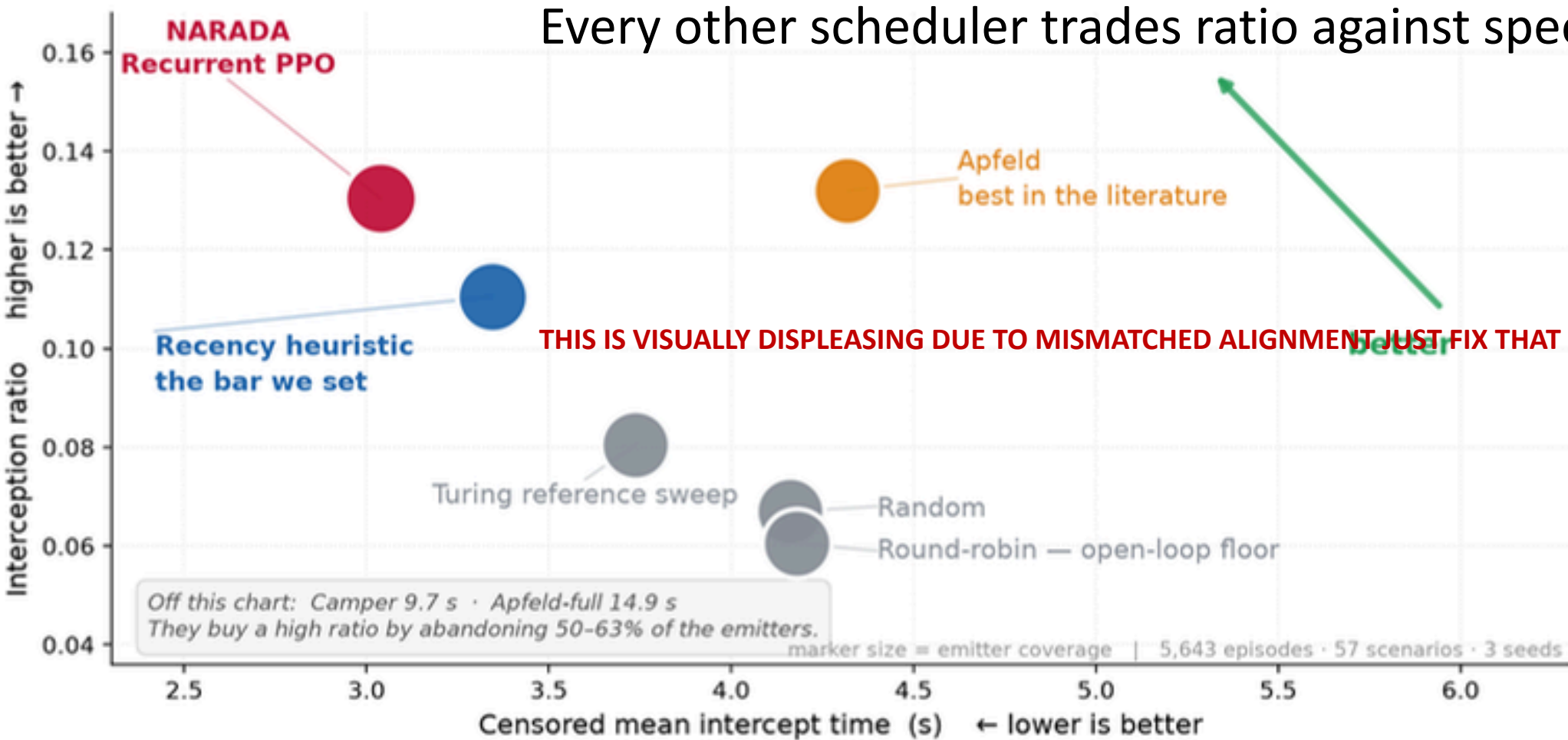
Play with it have fun
Every ES system already carries a threat library
NARADA's scheduler can take a priority input from the environment .

make this table pop out use vibrant colors

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NARADA'S USP

ADAPTIVE THREAT
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Four layers of linear screening so NARADA'S scheduler's optimizes the 3 most important metrics

Basically make this

01 Scenario

2.2 GB of real recorded radar data that gives 3,443 reusable emitter contributions. Every episode is fresh and new

02 Truth

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04 Scheduler

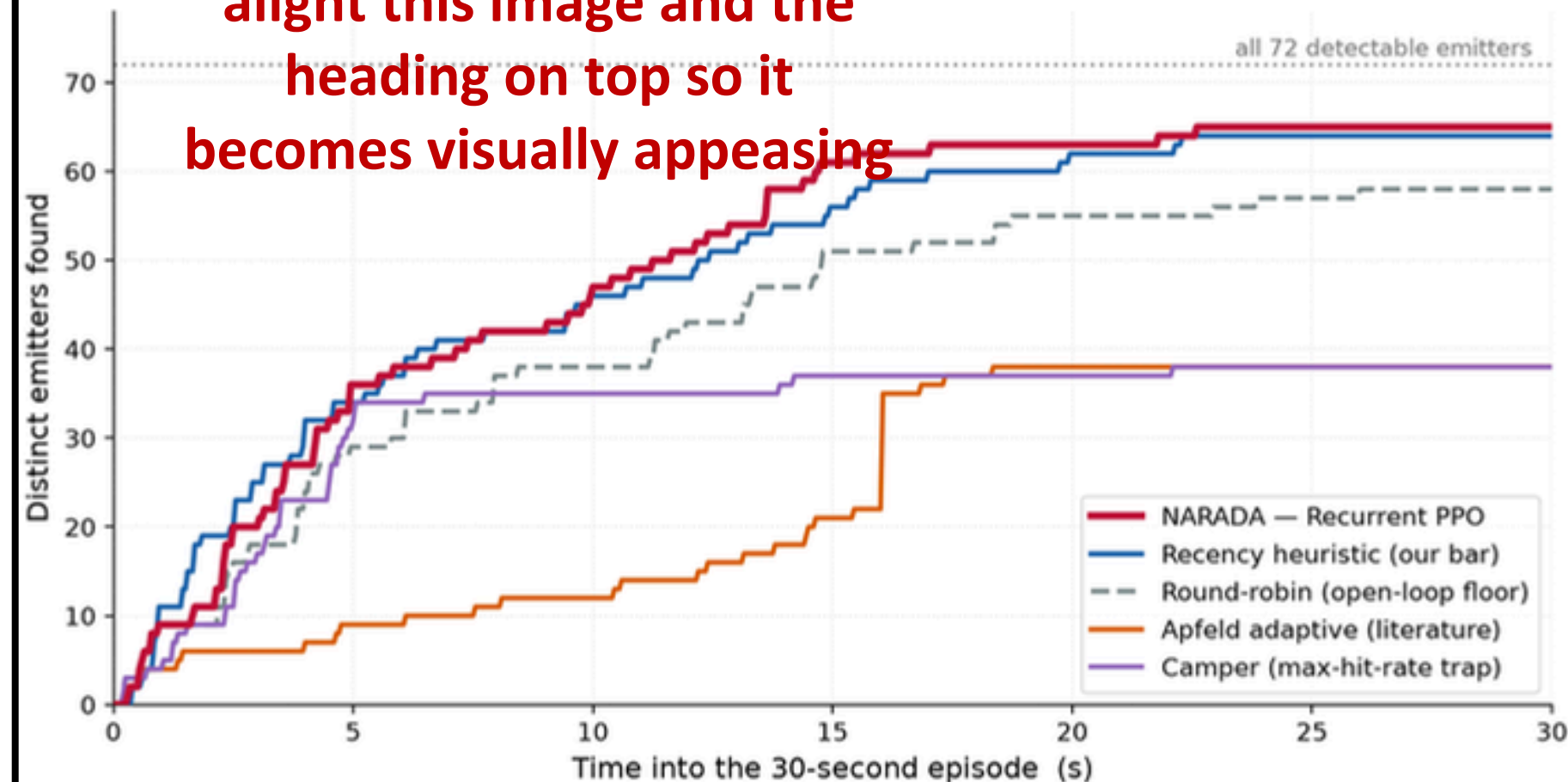
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align this image and the heading on top so it becomes visually appealing



Interception ratio

Intercept time

Coverage

We have to put how to improve the periodic tracking here

Reward = Exploit what pays + Explore what's overdue – Cost of hogging airspace.

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FEASIBILITY AND VIABILITY



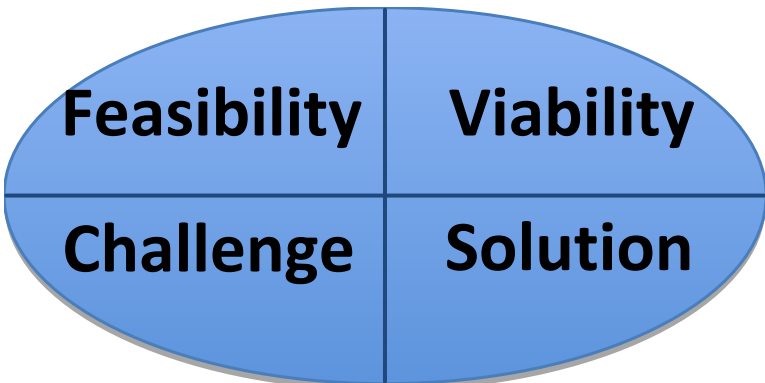
NARADA already runs feasibly on 5,643 episodes, 57 scenarios and 3 seeds.

NARADA has this right now

- Environment, 4 validation gates, 9-rung baseline ladder,A trained agent
- Runs on one CPU at inference and can make 600 decisions per 30-second episode.
- Matches the exact receiver geometry named in the PS's own dataset that is 36 bands and ± 500 MHz

NARADA can viably transfer it on the ground

- It Only reads hit/no-hit which is compatible every ES receiver that already produces that signal.
- Has a Human on the loop mot in it so it can work without human input whereas if there is valuable Human intelligence present it can be easily fed into the system
- We have kept 45 real recordings untouched for a clean final test.



- Should we Chase one metric alone to make the interception great? A "camper" agent scored high on hit-rate but ignored half the emitters.
- How to stop the agent from memorizing the training data and focus on learning? We caught this happening in our own work.
- How do we actually differentiate Random luck when it can look like a real result? our checkpoints varied 18.7-54.4% by chance alone.

- Every result reports three metrics together (Interception ratio, Interception time, Coverage) never just one.
- We removed any overlap between training and test data, enforced by an automated check.
- Averaged across three random seeds, with the winning model chosen by a rule set before we saw any scores.

POINTS SHOULD BE ALIGNED TO THE POINTS BESIDE THEM IN THE PARALLEL BOXES

Immediate impact on ES capacity and can be layered with EA & EP cognitive layers

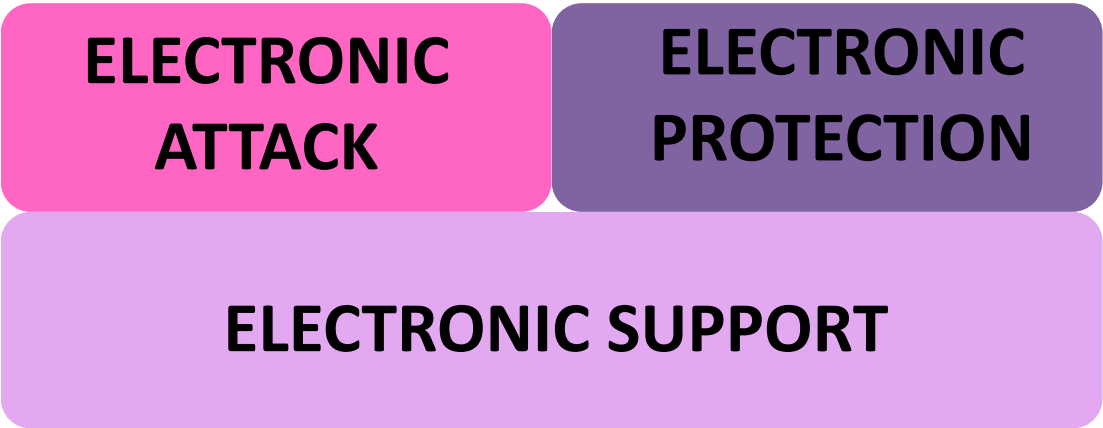
IMPACT

Can be directly integrated with already existing DRDO solutions like Legacy Samyukta and the new GBMES by DLRL as they both take binary YES/NO or Hit/Miss Input



In an Active warfare scenario where we are in hostile territory or our territory becomes hostile, patterns break the enemy uses systems that don't exist in our threat library, NARADA acts as a guide.

EASE OF LAYERING



BENIFITS AND FUTURE SCOPES

Increase the engagement capacity of SAM batteries



SAM's can track more enemies than they can engage smart scheduling can possibly improve that

Frequency Agile Response



Adaptive selection of jamming responses against frequency-agile emitters.

Reactive jamming



Future integration could enable prioritised responses based on threat relevance, timing and available resources.

Stay Operational Under Jamming



Help friendly systems keep working even when the enemy tries to jam them

Test Our Systems Against Similar Threats



Help friendly platforms avoid transmitting when it is not needed, making them harder to detect.

Train Our Systems Against Similar Threats



We use NARADA's logic to train our own systems could be detected or targeted by an enemy using a similar system and make them smarter.

THIS PAGE IS TOO CLUTTERED JUST CLEAN IT UP PLEASE PLEASEEEE

Them a 3 layer diagram that how easily it can be layerd with EA and ES

Your Team
Name

RESEARCH AND REFERENCES



References

Research

YOUTUBE VIDEO

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.